
Physics 1: Algebra-Based Practice Exam

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AP[®] PHYSICS 1 TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS

Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	Coulomb's law constant, $k = 1/4\pi\epsilon_0 = 9.0 \times 10^9$ N·m ² /C ²
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Universal gravitational constant, $G = 6.67 \times 10^{-11}$ m ³ /kg·s ²
Speed of light, $c = 3.00 \times 10^8$ m/s	Acceleration due to gravity at Earth's surface, $g = 9.8$ m/s ²

UNIT SYMBOLS	meter, m	kelvin, K	watt, W	degree Celsius, °C
	kilogram, kg	hertz, Hz	coulomb, C	
	second, s	newton, N	volt, V	
	ampere, A	joule, J	ohm, Ω	

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	1/2	3/5	$\sqrt{2}/2$	4/5	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	4/5	$\sqrt{2}/2$	3/5	1/2	0
$\tan \theta$	0	$\sqrt{3}/3$	3/4	1	4/3	$\sqrt{3}$	∞

The following conventions are used in this exam.

- I. The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- II. Assume air resistance is negligible unless otherwise stated.
- III. In all situations, positive work is defined as work done on a system.
- IV. The direction of current is conventional current: the direction in which positive charge would drift.
- V. Assume all batteries and meters are ideal unless otherwise stated.

AP[®] PHYSICS 1 EQUATIONS

MECHANICS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$$

$$|\vec{F}_f| \leq \mu |\vec{F}_n|$$

$$a_c = \frac{v^2}{r}$$

$$\vec{p} = m\vec{v}$$

$$\Delta \vec{p} = \vec{F} \Delta t$$

$$K = \frac{1}{2} m v^2$$

$$\Delta E = W = F_{\parallel} d = F d \cos \theta$$

$$P = \frac{\Delta E}{\Delta t}$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega = \omega_0 + \alpha t$$

$$x = A \cos(2\pi f t)$$

$$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$$

$$\tau = r_{\perp} F = r F \sin \theta$$

$$L = I \omega$$

$$\Delta L = \tau \Delta t$$

$$K = \frac{1}{2} I \omega^2$$

$$|\vec{F}_s| = k |\vec{x}|$$

$$U_s = \frac{1}{2} k x^2$$

$$\rho = \frac{m}{V}$$

a = acceleration
 A = amplitude
 d = distance
 E = energy
 f = frequency
 F = force
 I = rotational inertia
 K = kinetic energy
 k = spring constant
 L = angular momentum
 ℓ = length
 m = mass
 P = power
 p = momentum
 r = radius or separation
 T = period
 t = time
 U = potential energy
 V = volume
 v = speed
 W = work done on a system
 x = position
 y = height
 α = angular acceleration
 μ = coefficient of friction
 θ = angle
 ρ = density
 τ = torque
 ω = angular speed

$$\Delta U_g = m g \Delta y$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$|\vec{F}_g| = G \frac{m_1 m_2}{r^2}$$

$$\vec{g} = \frac{\vec{F}_g}{m}$$

$$U_G = -\frac{G m_1 m_2}{r}$$

ELECTRICITY

$$|\vec{F}_E| = k \left| \frac{q_1 q_2}{r^2} \right|$$

$$I = \frac{\Delta q}{\Delta t}$$

$$R = \frac{\rho \ell}{A}$$

$$I = \frac{\Delta V}{R}$$

$$P = I \Delta V$$

$$R_s = \sum_i R_i$$

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

A = area
 F = force
 I = current
 ℓ = length
 P = power
 q = charge
 R = resistance
 r = separation
 t = time
 V = electric potential
 ρ = resistivity

WAVES

$$\lambda = \frac{v}{f}$$

f = frequency
 v = speed
 λ = wavelength

GEOMETRY AND TRIGONOMETRY

Rectangle
 $A = bh$
 A = area
 C = circumference
 V = volume

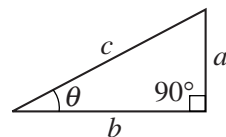
Triangle
 $A = \frac{1}{2} bh$
 S = surface area
 b = base
 h = height
 ℓ = length

Circle
 $A = \pi r^2$
 $C = 2\pi r$
 w = width
 r = radius

Rectangular solid
 $V = \ell wh$
Right triangle
 $c^2 = a^2 + b^2$

Cylinder
 $V = \pi r^2 \ell$
 $S = 2\pi r \ell + 2\pi r^2$
 $\sin \theta = \frac{a}{c}$
 $\cos \theta = \frac{b}{c}$
 $\tan \theta = \frac{a}{b}$

Sphere
 $V = \frac{4}{3} \pi r^3$
 $S = 4\pi r^2$



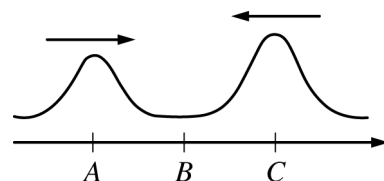
PHYSICS 1
Section I
40 Questions
Time—90 minutes

Note: To simplify calculations, you may use $g = 10 \text{ m/s}^2$ in all problems.

Directions: Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

1. A blue sphere and a red sphere with the same diameter are released from rest at the top of a ramp. The red sphere takes a longer time to reach the bottom of the ramp. The spheres are then rolled off a horizontal table at the same time with the same speed and fall freely to the floor. Which sphere reaches the floor first?

- (A) The red sphere
 (B) The blue sphere
 (C) The sphere with the greater mass
 (D) Neither; the spheres reach the floor at the same time.



	Amplitudes		
Time t	Point A	Point B	Point C
0 s	0.2 m	0	0.3 m
0.2 s	0	x	0
0.4 s	y	0	z

2. A 12 kg box sliding on a horizontal floor has an initial speed of 4.0 m/s. The coefficient of friction between the box and the floor is 0.20. The box moves a distance of 4.0 m in 2.0 s. The magnitude of the change in momentum of the box during this time is most nearly

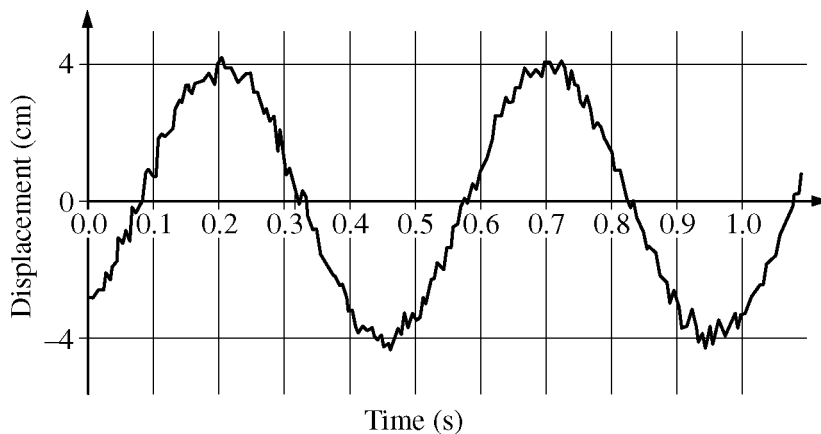
- (A) 12 kg•m/s
 (B) 48 kg•m/s
 (C) 60 kg•m/s
 (D) 96 kg•m/s

3. The figure above shows two wave pulses on a spring traveling toward each other at time $t = 0$. At $t = 0.2$ s the pulses completely overlap. A student measures the amplitudes at points A, B, and C at the three times shown in the table. Which of the following is the most likely set of values that the student recorded for x , y , and z in the table?

	x	y	z
(A)	0.5 m	0.2 m	0.3 m
(B)	0.5 m	0.3 m	0.2 m
(C)	0.1 m	0.2 m	0.3 m
(D)	0.1 m	0.3 m	0.2 m

4. A rubber ball with mass 0.20 kg is dropped vertically from a height of 1.5 m above a floor. The ball bounces off of the floor, and during the bounce 0.60 J of energy is dissipated. What is the maximum height of the ball after the bounce?
- (A) 0.30 m
(B) 0.90 m
(C) 1.2 m
(D) 1.5 m
5. A sled slides down a hill with friction between the sled and hill but negligible air resistance. Which of the following must be correct about the resulting change in energy of the sled-Earth system?
- (A) The sum of the kinetic energy and the gravitational potential energy changes by an amount equal to the energy dissipated by friction.
(B) The gravitational potential energy decreases and the kinetic energy is constant.
(C) The decrease in the gravitational potential energy is equal to the increase in kinetic energy.
(D) The gravitational potential energy and the kinetic energy must both decrease.

Questions 6-8 refer to the following material.

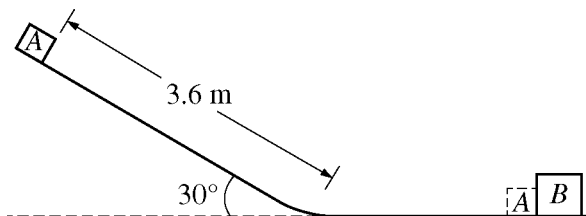


A student sets an object attached to a spring into oscillatory motion and uses a position sensor to record the displacement of the object from equilibrium as a function of time. A portion of the recorded data is shown in the figure above.

6. The speed of the object at time $t = 0.65$ s is most nearly equal to which of the following?
- (A) The value of the graph at 0.65 s
 - (B) The slope of the line connecting the origin and the point on the graph at 0.65 s
 - (C) The slope of the line connecting the point where the graph crosses the time axis near 0.57 s and the point on the graph at 0.65 s
 - (D) The slope of the tangent to a best-fit sinusoidal curve at 0.65 s
7. The total distance traveled by the object between 0.35 s and 0.40 s is most nearly
- (A) 0 cm
 - (B) 2 cm
 - (C) 4 cm
 - (D) 6 cm
8. The frequency of oscillation is most nearly
- (A) 0.5 Hz
 - (B) 0.7 Hz
 - (C) 1.4 Hz
 - (D) 2.0 Hz

9. An object is moving to the west at a constant speed. Three forces are exerted on the object. One force is 10 N directed due north, and another is 10 N directed due west. What is the magnitude and direction of the third force if the object is to continue moving to the west at a constant speed?

(A) $10\sqrt{3}$ N , directed northwest
(B) $10\sqrt{3}$ N , directed southeast
(C) $10\sqrt{2}$ N , directed northwest
(D) $10\sqrt{2}$ N , directed southeast



10. Block A of mass 2.0 kg is released from rest at the top of a 3.6 m long plane inclined at an angle of 30°, as shown in the figure above. After sliding on the horizontal surface, block A hits and sticks to block B, which is at rest and has mass 3.0 kg. Assume friction is negligible. The speed of the blocks after the collision is most nearly
- (A) 2.4 m/s
(B) 3.2 m/s
(C) 3.8 m/s
(D) 6.0 m/s

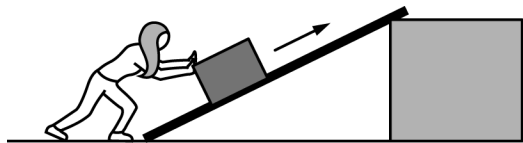


Figure 1

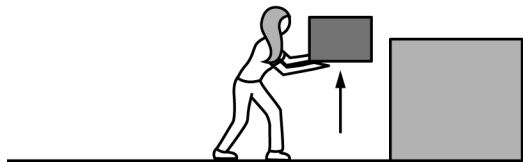
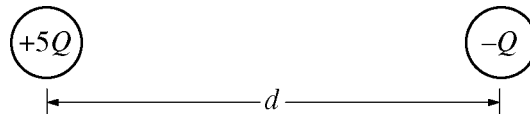


Figure 2

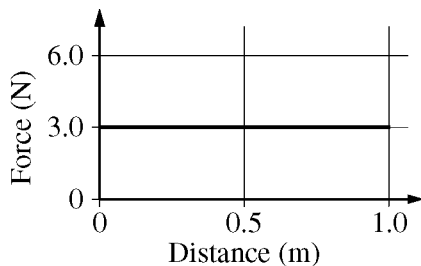
11. A student is asked to move a box from ground level to the top of a loading dock platform, as shown in the figures above. In Figure 1, the student pushes the box up an incline with negligible friction. In Figure 2, the student lifts the box straight up from ground level to the loading dock platform. In which case does the student do more work on the box, and why?
- (A) Lifting the box straight up, because it requires a larger applied force to lift it straight up
 - (B) Pushing the box up the incline, because the force is applied for a longer distance
 - (C) Lifting the box straight up, because the incline acts as a simple machine and reduces the force required
 - (D) Neither method, because the work is the same in both cases, since using the ramp decreases the force by the same factor that it increases the distance



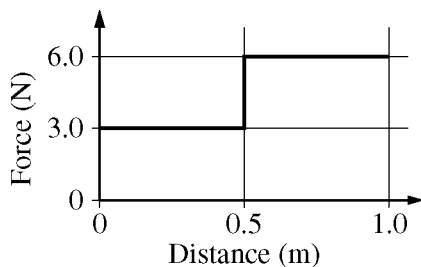
12. Two charged conducting spheres are separated by a distance d , as shown in the figure above. One sphere has charge $+5Q$, the other has charge $-Q$, and the diameter of each sphere is much smaller than d . Each sphere exerts a force of magnitude F on the other. The spheres are brought together and touch briefly. After touching, the left sphere has charge $+2Q$. The spheres are then returned to their original locations. What is the magnitude of the new force between the spheres?
- (A) $\frac{2}{5}F$
 - (B) $\frac{4}{5}F$
 - (C) $\frac{9}{5}F$
 - (D) $4F$
13. The wheel on a vehicle has a rotational inertia of $2.0 \text{ kg}\cdot\text{m}^2$. At the instant the wheel has a counterclockwise angular velocity of 6.0 rad/s , an average counterclockwise torque of $5.0 \text{ N}\cdot\text{m}$ is applied, and continues for 4.0 s . What is the change in angular momentum of the wheel?
- (A) $12 \text{ kg}\cdot\text{m}^2/\text{s}$
 - (B) $16 \text{ kg}\cdot\text{m}^2/\text{s}$
 - (C) $20 \text{ kg}\cdot\text{m}^2/\text{s}$
 - (D) $32 \text{ kg}\cdot\text{m}^2/\text{s}$

14. A student uses a spring scale to exert a horizontal force on a block, pulling the block over a smooth floor. The student repeats the procedure several times, each time pulling the block from rest through a distance of 1.0 m. For which of the following graphs of force as a function of distance will the block be moving the fastest at the end of the 1.0 m?

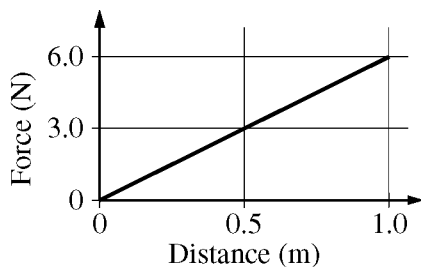
(A)



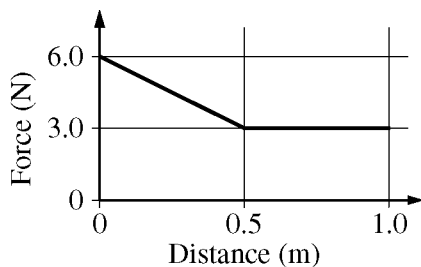
(B)



(C)



(D)

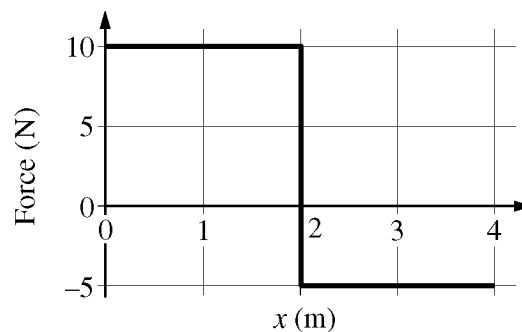
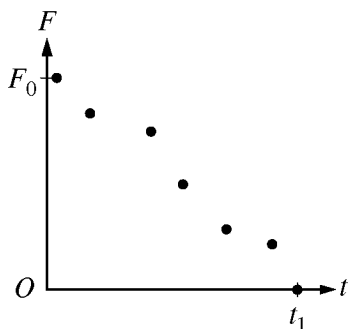


15. A projectile fired into the air explodes and splits into two halves of equal mass that hit the ground at the same time. If the projectile had not exploded, it would have landed at point X , which is a distance R to the right of the launch point. After the explosion, one of the halves lands at point Y , which is a distance $2R$ to the right of the launch point. If air resistance is negligible, where did the other half land?

- (A) To the left of the launch point
(B) At the launch point
(C) Between the launch point and point X
(D) Between points X and Y

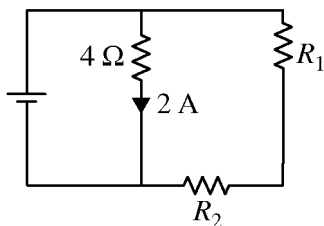
16. An elevator carrying a person of mass m is moving upward and slowing down. How does the magnitude F of the force exerted on the person by the elevator floor compare with the magnitude mg of the gravitational force?

- (A) $F < mg$
(B) $F = mg$
(C) $F > mg$
(D) F can be greater than or less than mg , depending on the speed of the elevator.



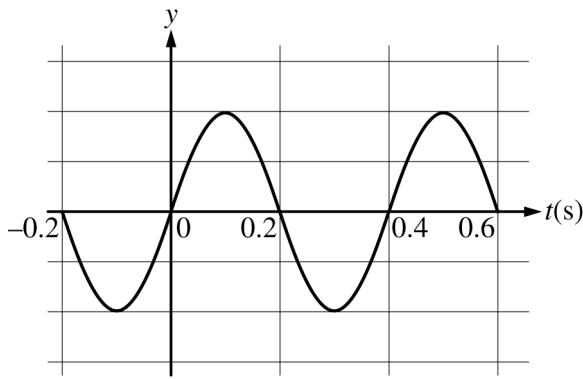
17. A soft foam block of mass m slides without friction in the positive x -direction with speed v . At time $t = 0$, a student briefly pushes the block with a force probe in the positive x -direction. The graph above shows the force probe's measurements as a function of time during the push. Which of the following statements is true about the block's momentum between $t = 0$ and $t = t_1$?
- (A) The momentum of the block has decreased to zero at time t_1 .
- (B) The momentum of the block has increased by approximately $\frac{1}{2} F_0 t_1$.
- (C) The momentum of the block has decreased by approximately $\frac{1}{2} F_0 t_1$.
- (D) The change in momentum cannot be determined without knowing the distance by which the force probe compressed the block.
18. A cart is moving on a level track in the positive x -direction. A force acting parallel to the x -axis is exerted on the cart. The graph above shows the net force exerted on the cart as a function of displacement. As the cart travels from $x = 0$ m to $x = 4$ m, what is the net change in the kinetic energy of the cart?
- (A) An increase of 20 J
- (B) An increase of 10 J
- (C) A decrease of 20 J
- (D) A decrease of 10 J

Questions 19-20 refer to the following material.



In the circuit shown above, the sum of the resistances of resistors R_1 and R_2 is $16\ \Omega$.

19. What is the current in the right branch of the circuit?
- (A) 0.25 A
(B) 0.50 A
(C) 1.00 A
(D) 2.00 A
20. Resistor R_1 and the $4\ \Omega$ resistor are now swapped. How does the current through the battery change, and why?
- (A) The current does not change, because the resistance does not change.
(B) The current increases, because the total resistance will always decrease.
(C) The current decreases, because the total resistance will always increase.
(D) The change in current cannot be determined without knowing the resistances of R_1 and R_2 .

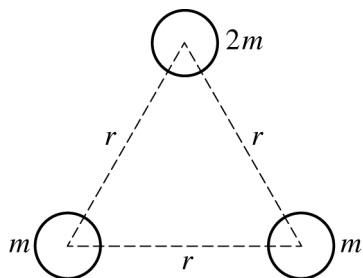


21. A transverse periodic wave travels along a string that is stretched along the x -axis. The figure above shows the y -position of a point on the string as a function of time. What further information, if any, is needed to determine the wavelength of the wave?

- (A) The velocity of the wave
- (B) The amplitude of the wave
- (C) The frequency of the wave
- (D) No further information is needed to determine the wavelength.

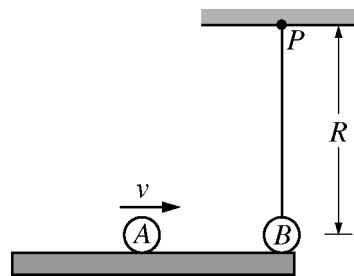
22. Two objects, A and B , move toward one another. Object A has twice the mass and half the speed of object B . Which of the following describes the forces the objects exert on each other when they collide and provides the best explanation?

- (A) The force exerted by A on B will be twice as great as the force exerted by B on A , because A has twice the mass of B .
- (B) The force exerted by A on B will be half as great as the force exerted by B on A , because A has half the speed of B .
- (C) The forces exerted by each object on the other are the same, because the product of mass and speed is the same for both objects.
- (D) The forces exerted by each object on the other are the same, because interacting objects cannot exert forces of different magnitude on each other.



23. Three spheres, with masses indicated above, are initially far away from each other, and the gravitational potential energy of the three-sphere system is zero. The spheres are then brought together until each sphere is a distance r from the other two, as shown above. What is the new gravitational potential energy of the three-sphere system?

- (A) $-\frac{Gm^2}{r}$
 (B) $-\frac{2Gm^2}{r}$
 (C) $-\frac{4Gm^2}{r}$
 (D) $-\frac{5Gm^2}{r}$



24. Steel sphere A of mass M is moving along a horizontal surface with constant speed v . Identical steel sphere B is at rest and hangs on a string of length R attached to a support at point P , as shown in the figure above. The spheres collide, and as a result sphere A stops and sphere B swings a vertical height h before coming momentarily to rest. Knowing values for which of the following will allow determination of the angular impulse on sphere B with respect to P due to the collision?

- (A) M and v only
 (B) M , v , and h
 (C) R and h
 (D) R , M , and v

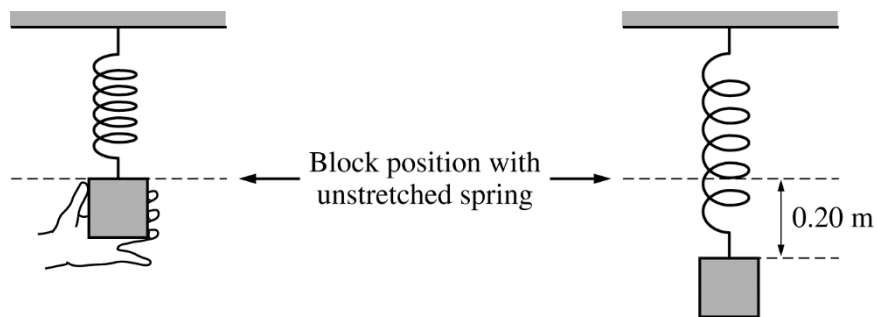
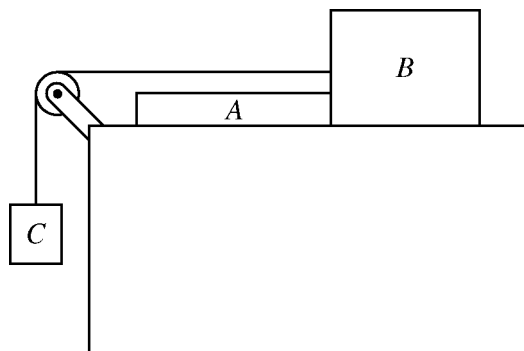


Figure 1

Figure 2

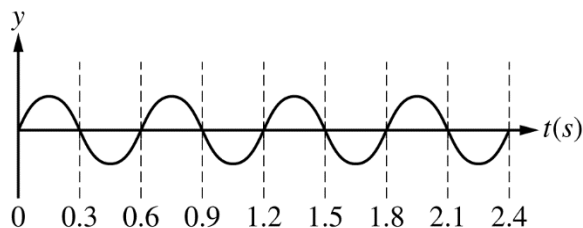
25. A 1.0 kg block is attached to an unstretched spring of spring constant 50 N/m and released from rest from the position shown in Figure 1 above. The block oscillates for a while and eventually stops moving 0.20 m below its starting point, as shown in Figure 2. What is the change in potential energy of the block-spring-Earth system between Figure 1 and Figure 2 ?

- (A) -1.0 J
- (B) 0 J
- (C) 1.0 J
- (D) 3.0 J



26. Blocks A and B , of masses m_A and m_B , are at rest on a frictionless surface, as shown above, with block A fixed to the table. Block C of mass m_C is suspended by a string that is tied to block B over an ideal pulley. Which of the following gives the magnitude of the force exerted by block A on block B ?

- (A) $m_B g$
- (B) $m_C g$
- (C) $\frac{m_A m_C}{m_A + m_B} g$
- (D) $\frac{m_B m_C}{m_A + m_B} g$

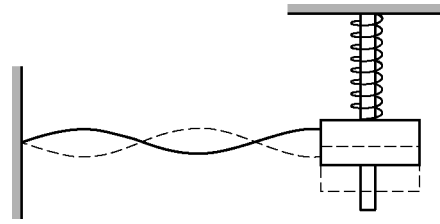


27. A student shakes a horizontally-stretched cord, creating waves. The graph above shows the vertical position y as a function of time t for a point on the cord. The student then tightens the cord so that waves on it will travel faster than before. How should the student now shake the cord to make the graph of y versus t for the point look the same as above?
- (A) With fewer shakes per second than before
 - (B) With the same number of shakes per second as before
 - (C) With more shakes per second than before
 - (D) The answer cannot be determined without knowing the wavelength of the waves.

28. A cart of mass m is moving with negligible friction along a track with known speed v_1 to the right. It collides with and sticks to a cart of mass $4m$ moving with known speed v_2 to the right. Which of the two principles, conservation of momentum and conservation of mechanical energy, must be applied to determine the final speed of the carts, and why?

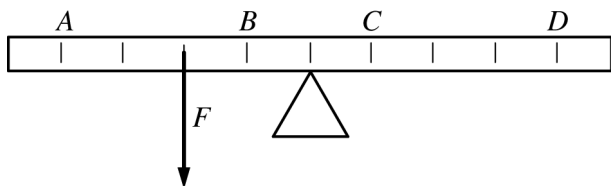
- (A) Only conservation of momentum, because the momentum lost by one cart is gained by the other and there is only one unknown quantity.
- (B) Both conservation of mechanical energy and conservation of momentum, because both principles apply in any collision.
- (C) Both conservation of mechanical energy and conservation of momentum, because neither cart changes direction.
- (D) Either conservation of momentum or conservation of mechanical energy, because only one equation is required to solve for the one unknown variable.

Questions 29-30 refer to the following material.



The figure above shows a pole with a spring around it and a 2.5 kg block with a hole in the middle hanging from the spring. A light horizontal cord is attached to the block and a wall. The block is oscillating at 10.0 Hz, and the standing wave shown is formed.

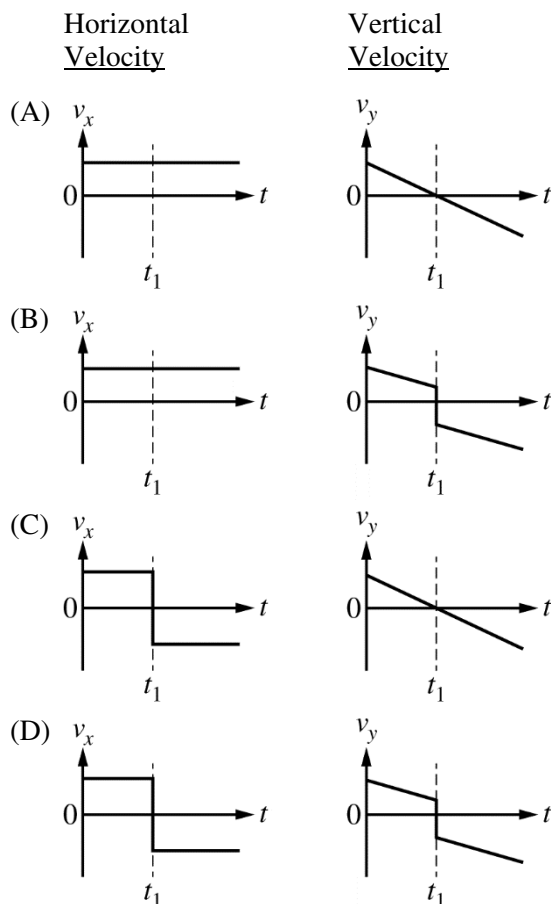
29. The spring constant of the spring is approximately which of the following?
- (A) 10 N/m
 - (B) 100 N/m
 - (C) 1×10^3 N/m
 - (D) 1×10^4 N/m
30. What additional measurement is needed to determine the speed of the wave on the cord?
- (A) The thickness of the cord
 - (B) The unstretched length of the spring
 - (C) The average kinetic energy of the block
 - (D) The distance between nodes on the cord



31. The figure above shows a uniform meterstick that is set on a fulcrum at its center. A force of magnitude F toward the bottom of the page is exerted on the meterstick at the position shown. At which of the labeled positions must an upward force of magnitude $2F$ be exerted on the meterstick to keep the meterstick in equilibrium?

(A) A
(B) B
(C) C
(D) D

32. In a classroom at time $t = 0$, a sphere is thrown upward at a 45° angle to the horizontal. At time t_1 , while the sphere is still rising, it bounces off the ceiling elastically and with no friction. Which of the following pairs of graphs could represent the sphere's horizontal velocity and vertical velocity as functions of time t ?



33. Two satellites are in circular orbits around Earth.

Satellite A has speed v_A . Satellite B has an orbital radius nine times that of satellite A .

What is the speed of satellite B ?

- (A) $v_A/9$
- (B) $v_A/3$
- (C) $3v_A$
- (D) $9v_A$

34. In trial 1 of an experiment, a cart moves with speed v_0 on a frictionless, horizontal track and collides elastically with another cart that is initially at rest. In trial 2, the setup is identical except that the carts stick together during the collision. How does the speed of the two-cart system's center of mass change, if at all, during the collision in each trial?

Trial 1

Trial 2

- | | |
|---------------------|-----------------|
| (A) Does not change | Does not change |
| (B) Does not change | Decreases |
| (C) Decreases | Does not change |
| (D) Decreases | Decreases |

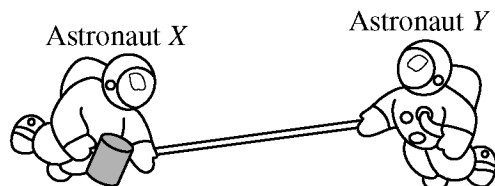


Figure 1

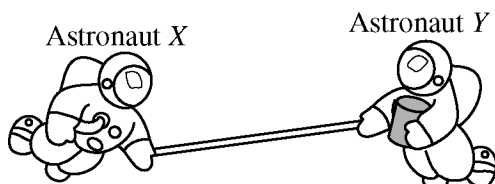


Figure 2

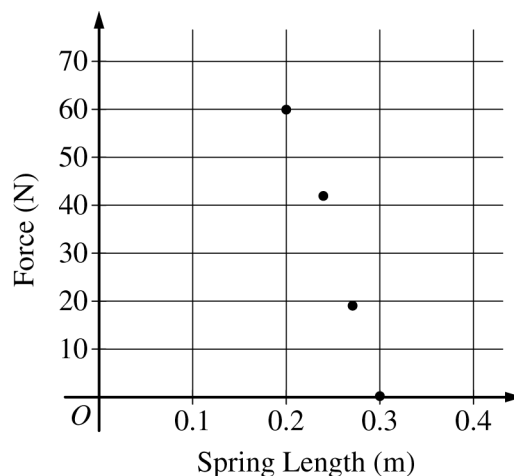
35. Two astronauts are connected by a taut cable and are initially at rest with respect to a nearby space station. Astronaut X throws a large container to Astronaut Y. Figure 1 above shows the astronauts immediately after the container is thrown by Astronaut X, and Figure 2 shows the astronauts immediately after the container is caught by Astronaut Y. Which of the following describes the motion of Astronaut Y in Figures 1 and 2?

Figure 1

- (A) Does not move
- (B) Moves to the left
- (C) Does not move
- (D) Moves to the left

Figure 2

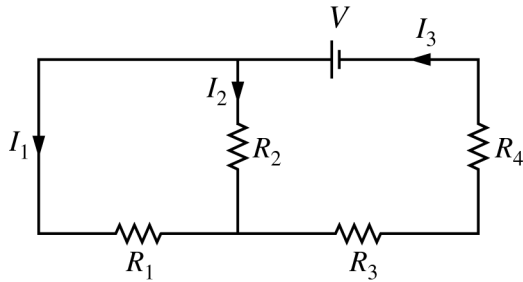
- Moves to the right
- Moves to the right
- Does not move
- Does not move



36. The graph above shows the force exerted by a spring as a function of the length of the spring. A block on a frictionless table is pushed against the spring that is fastened to a wall. The spring is compressed until its length is 20 cm. The block is then released. Which of the following values is closest to the kinetic energy with which the block leaves the spring?

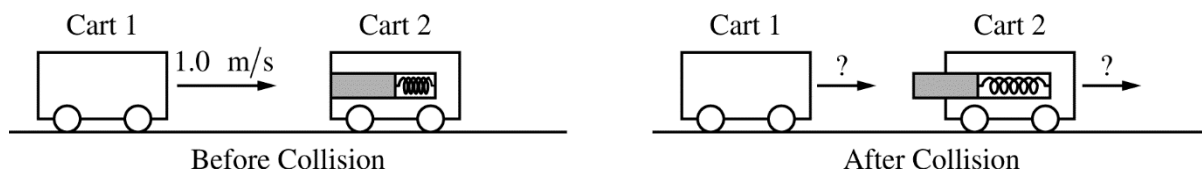
- (A) 3 J
- (B) 6 J
- (C) 12 J
- (D) 15 J

Directions: For each of the questions or incomplete statements below, two of the suggested answers will be correct. For each of these questions, you must select both correct choices to earn credit. No partial credit will be earned if only one correct choice is selected. Select the two that are best in each case and then fill in the corresponding circles that begin with number 131 on page 3 of the answer sheet.



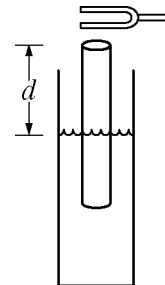
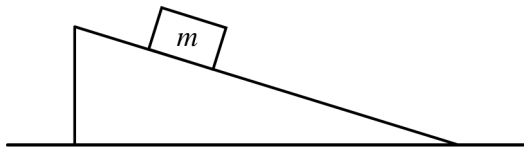
131. Which of the following equations correctly represent the total potential difference for a complete loop in the circuit diagram above? Select two answers.

- (A) $V - I_1 R_1 - I_3 (R_3 + R_4) = 0$
- (B) $V - I_1 R_1 - I_2 R_2 - I_3 (R_3 + R_4) = 0$
- (C) $V - I_3 (R_3 + R_4) - I_2 R_2 = 0$
- (D) $V - I_1 R_1 + I_2 R_2 = 0$



132. Two lab carts have the same mass and are free to move on a horizontal track. The carts' wheels have negligible mass. Cart 1 travels to the right at 1.0 m/s and collides with cart 2, which is initially at rest, as shown at left above. Cart 2 has a compressed spring-loaded plunger with a nonnegligible amount of stored energy. During the collision, the spring-loaded plunger pops out, staying in contact with cart 1 for 0.10 s as the spring decompresses. Negligible mechanical energy dissipates during the collision. Taking rightward as positive, the carts' velocities after the collision could be which of the following? Select two answers.

<u>Cart 1</u>	<u>Cart 2</u>
(A) 0	1.0 m/s
(B) 0.5 m/s	0.5 m/s
(C) -0.5 m/s	1.5 m/s
(D) -1.0 m/s	2.0 m/s

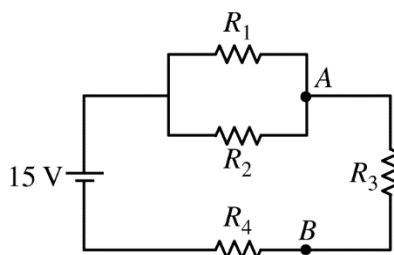


133. A block of mass m is at rest on a rough incline, as shown in the figure above. Which of the following forces must have a magnitude equal to mg ? Select two answers.
- (A) The total force exerted on the block by the incline
 - (B) The normal force exerted on the block by the incline
 - (C) The force of friction exerted on the block by the incline
 - (D) The gravitational force exerted on Earth by the block

134. A tuning fork is held above the opening of a pipe that is partially submerged in water, as shown in the figure above. The tuning fork is vibrating at known frequency f . Which of the following additional pieces of information, by itself, would allow the speed of sound in air to be determined? Select two answers.
- (A) The shortest air column length that gives a resonance
 - (B) The longest air column length that gives a resonance
 - (C) The difference between air column lengths that give two consecutive resonances
 - (D) The average of the air column lengths that give two consecutive resonances

PHYSICS 1
Section II
5 Questions
Time—90 minutes

Directions: Questions 1, 4, and 5 are short free-response questions that require about 13 minutes each to answer and are worth 7 points each. Questions 2 and 3 are long free-response questions that require about 25 minutes each to answer and are worth 12 points each. Show your work for each part in the space provided after that part.



1. (7 points, suggested time 13 minutes)

A student assembles a circuit with four resistors connected to a 15 V battery, as shown above. Let I_1 be the current through resistor R_1 , I_2 be the current through resistor R_2 , and so on. The ranking of the resistances of the resistors is $R_4 > R_3 > R_2 > R_1$.

- (a) Rank the four currents I_1 through I_4 in order from greatest to least. If any of the currents are the same, indicate that explicitly.

Ranking:

Briefly explain your reasoning.

- (b) The student connects a separate voltmeter across each resistor. Let ΔV_1 , ΔV_2 , ΔV_3 , and ΔV_4 be the voltmeter readings across resistors R_1 , R_2 , R_3 , and R_4 , respectively. The student measures $\Delta V_1 = 1 \text{ V}$ and $\Delta V_4 = 9 \text{ V}$. Based on these measurements, determine the voltmeter readings ΔV_2 and ΔV_3 .

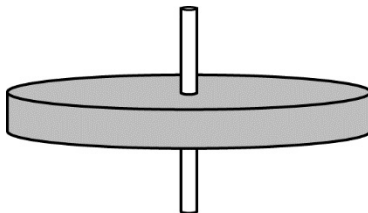
$$\Delta V_2 = \underline{\hspace{2cm}} \quad \Delta V_3 = \underline{\hspace{2cm}}$$

Briefly explain your reasoning.

- (c) The student now connects a wire from point A to point B . When this wire is connected, does the current I_1 in resistor R_1 increase, decrease, or stay the same?

☐ Increase ☐ Decrease ☐ Stay the same

Briefly explain your reasoning.



2. (12 points, suggested time 25 minutes)

A disk-shaped platform has a known rotational inertia. The platform is mounted on a fixed axle and rotates in a horizontal plane, as shown above. A student wishes to determine the frictional torque exerted on the platform by the axle as the platform rotates. The student has access to equipment that would usually be found in a school physics laboratory.

- (a) Describe an experimental procedure the student could use to collect the data needed to find the frictional torque exerted on the platform while it rotates.
- What quantities would be measured?
 - What equipment would be used for the measurements, and how would that equipment be used?
 - Describe the overall procedure to be used, including any steps necessary to reduce experimental uncertainty. Give enough detail so that another student could replicate the experiment.

- (b) Describe how the data from the measurements could be analyzed to determine the frictional torque exerted on the rotating platform.

It is often assumed that the frictional force between two surfaces is independent of their relative speed. However, the details of the axle's construction are unknown, and it is possible that the frictional torque for the axle depends on the platform's angular speed.

- (c) Does the experiment described in parts (a) and (b) depend on the assumption that the frictional torque is independent of the platform's angular speed?

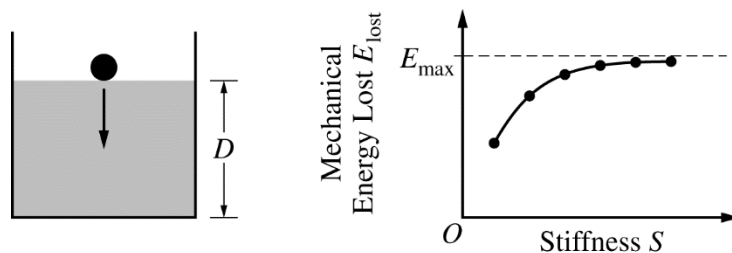
_____ Yes _____ No

Note: Either answer may be correct, depending on your experimental design.

If you answered yes, describe which part of your analysis depends on this assumption.

If you answered no, explain how your analysis does not depend on this assumption.

- (d) Briefly describe how the experimental procedure described in parts (a) and (b) could be modified to determine whether the frictional torque stays constant as the angular speed changes. If no modification is necessary, state that explicitly. In either case, describe how the data could be analyzed to determine whether the frictional torque stays constant as the angular speed changes.



3. (12 points, suggested time 25 minutes)

Food scientists have created a new oil. At room temperature, the oil is a liquid. As the oil gets colder, however, it stiffens (thickens) into a sticky gel.

To explore the properties of the oil, the scientists fill a container with the oil to a height D , as shown in the figure above on the left. They drop a small steel ball of mass M from rest at the top of the oil. Using video to capture the ball's motion, the scientists calculate E_{lost} , the mechanical energy lost by the ball-Earth system from the time the ball enters the oil to the time just before the ball strikes the bottom of the container.

The scientists also define the “stiffness” S of the oil as a quantity proportional to the force required to move a rod through the oil at a standard constant speed. The scientists calculate E_{lost} and S at several different temperatures, ranging from room temperature to the lowest temperature at which the ball still falls through the oil. The graph above on the right shows E_{lost} as a function of S for the calculated data points and a best-fit curve.

- (a) Give a physical reason why the curve in the graph would not reach the vertical (E_{lost}) axis even if the scientists had taken data over a broader range of temperatures.
- (b) As S increases, E_{lost} approaches a maximum value labeled E_{max} on the graph above. Write an equation for E_{max} in terms of M , D , and physical constants, as appropriate.

Justify your answer.

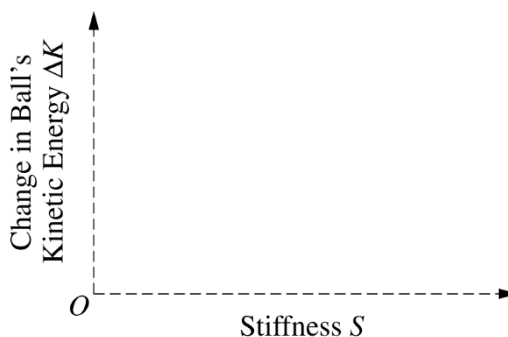
- (c) One of the scientists, in trying to represent the relationship between the oil stiffness and the mechanical energy lost, writes down the equation $E_{\text{lost}} = CS^2$, where C is a constant with appropriate units. Another scientist points out that this equation cannot be correct. Give two reasons why the equation cannot be correct.

- (d) Further attempting to model the ball's motion, the scientists write the following equation for the time t the ball takes to fall through the oil: $t = \frac{Z}{S}$, where Z is a constant with appropriate units. Is this equation plausible—in other words, does it make physical sense?

_____ Plausible _____ Not plausible

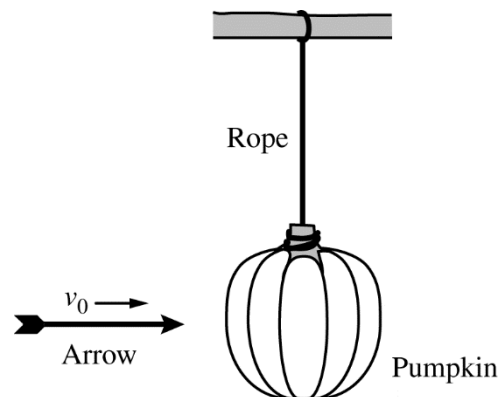
Briefly explain your reasoning.

- (e) ΔK is the change in kinetic energy of the ball between the time it is released from rest and the time just before the ball strikes the bottom of the container. On the axes below, sketch ΔK as a function of S , the oil stiffness.



4. (7 points, suggested time 13 minutes)

An archer tests various arrowheads by shooting arrows at a pumpkin that is suspended from a tree branch by a rope, as shown to the right. When struck head-on by the arrow, the pumpkin swings upward on the rope. The maximum angle θ that the rope makes with the vertical is different for each arrowhead that the archer tests. Each arrow, including its arrowhead, has the same mass m and is shot with the same velocity v_0 toward the right. The arrowheads are made of different materials, however, and each behaves differently when it strikes the pumpkin, as described below.



- *Embedded arrow*: Strikes the pumpkin and remains embedded, while the pumpkin swings to angle θ_{emb} .
 - *Pass arrow*: Passes all the way through the pumpkin and continues traveling away from the archer, while the pumpkin swings to angle θ_{pass} .
 - *Bounce arrow*: Bounces off the pumpkin back toward the archer, while the pumpkin swings to angle θ_{bounce} .
- (a) Rank the three angles θ_{emb} , θ_{pass} , and θ_{bounce} from greatest to least in the spaces indicated below. Use “1” for the greatest angle, “2” for the next greatest, and so on. If any two or all three angles are the same, use the same number for their ranking.

_____ θ_{emb} _____ θ_{pass} _____ θ_{bounce}

- (b) In a clear, coherent, paragraph-length response that may also contain figures and/or equations, justify your ranking.

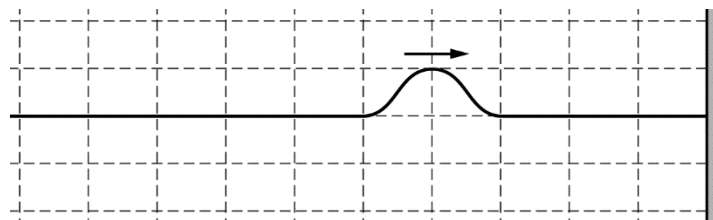


Figure 1. Pulse before reaching wall

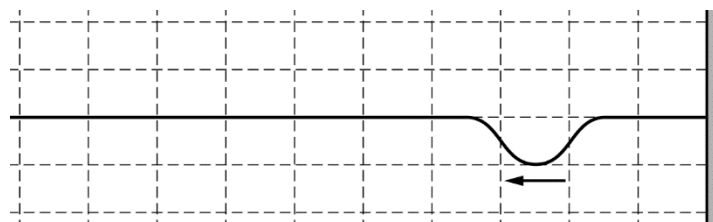
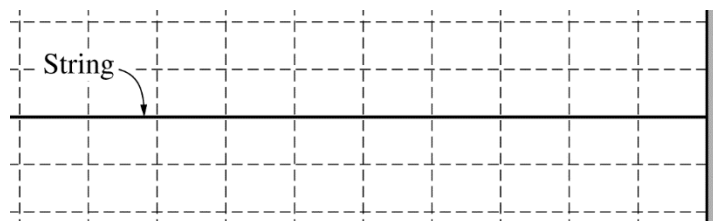


Figure 2. Pulse after reflecting from wall

5. (7 points, suggested time 13 minutes)

The right end of a long, flexible string is attached to a fixed point on a wall. The left end of the string is held by a student. The student creates a symmetric wave pulse traveling to the right, as shown in Figure 1 above. When the pulse reaches the wall, it inverts as it reflects from the wall, as shown in Figure 2 above. The figure below represents what the string looks like at the instant the peak (midpoint) of the pulse reaches the wall; at this instant the string is horizontal everywhere.



(a) At the instant shown above, does the wave pulse carry energy?

____ Yes ____ No

Briefly explain your answer.

Question 5 continues on the next page.

- (b) The student then creates a second symmetric wave pulse traveling to the right on the same string. The amplitude of the second pulse is half the amplitude of the original pulse. Which pulse, if either, travels at a higher speed?

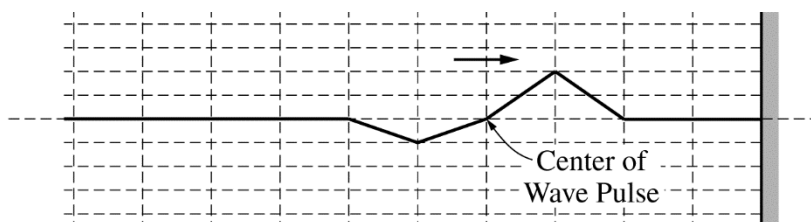
_____ The higher-amplitude pulse

_____ The lower-amplitude pulse

_____ Neither; both pulses travel at the same speed.

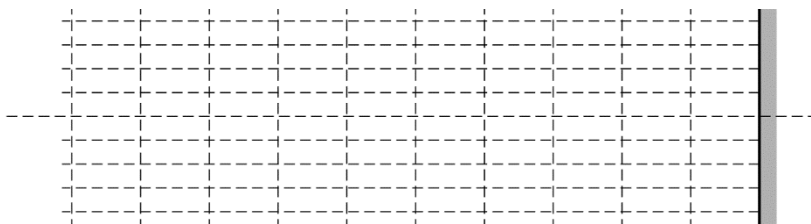
Briefly explain your reasoning.

- (c) The figure below shows an idealized wave pulse traveling to the right.



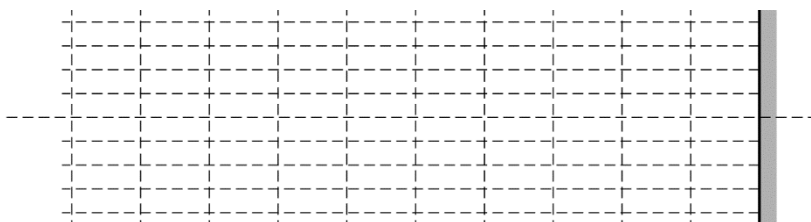
- i. Consider the instant when the center of the pulse reaches the wall, which is also the instant when the right half of the pulse has reflected from the wall but the left half has not. On the grid below, sketch what the string looks like at that instant.

Note: Do any scratch (practice) work on the grids on the following page. You will be graded only for the sketch made on the grid below.

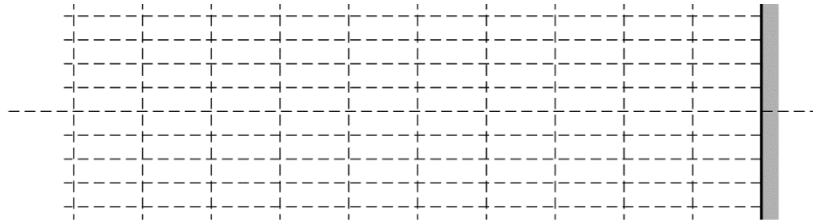
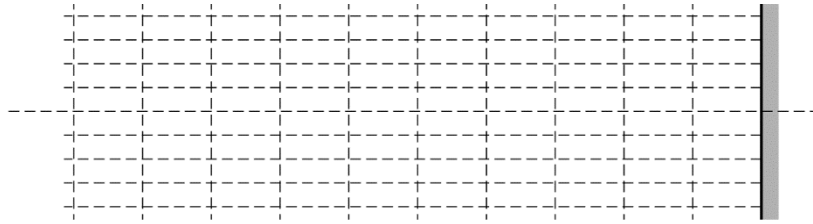


- ii. Now consider an instant sometime after the entire pulse has reflected from the wall and is moving to the left. On the grid below, sketch what the string looks like at that instant.

Note: Do any scratch (practice) work on the grids on the following page. You will be graded only for the sketch made on the grid below.



The grids below are provided for scratch work only. Sketches made below will NOT be graded.



Answer Key for AP Physics 1 Practice Exam, Section I

Question 1: D	Question 21: A
Question 2: B	Question 22: D
Question 3: B	Question 23: D
Question 4: C	Question 24: D
Question 5: A	Question 25: A
Question 6: D	Question 26: B
Question 7: B	Question 27: B
Question 8: D	Question 28: A
Question 9: D	Question 29: D
Question 10: A	Question 30: D
Question 11: D	Question 31: B
Question 12: B	Question 32: B
Question 13: C	Question 33: B
Question 14: B	Question 34: A
Question 15: B	Question 35: D
Question 16: A	Question 36: A
Question 17: B	Question 131: A, C
Question 18: B	Question 132: C, D
Question 19: B	Question 133: A, D
Question 20: D	Question 134: A, C

2017 AP Physics 1: Algebra-Based

Question Descriptors and Performance Data

Multiple-Choice Questions

Question	Learning Objectives	Essential Knowledge	Science Practices	Key	% Correct
1	3.A.1.1; 4.A.2.1	3.A.1; 4.A.2	1.5; 2.2; 6.4	D	74
2	3.A.1.1; 3.B.2.1; 4.B.2.1	3.A.1; 3.B.2; 4.B.2	1.5; 2.2; 1.1; 1.4; 2.2; 2.2	B	62
3	6.D.1.1; 6.D.1.2	6.D.1	1.4; 5.1	B	70
4	4.C.1.1	4.C.1	1.4; 2.2	C	66
5	5.B.5.4	5.B.5	6.4; 7.2	A	58
6	3.A.1.1; 3.A.1.3; 3.B.3.3	3.A.1; 3.B.3	1.5; 2.2; 5.1; 2.2; 5.1	D	49
7	3.A.1.1; 3.A.1.3	3.A.1	1.5; 2.2; 5.1	B	78
8	3.B.3.3	3.B.3	2.2; 5.1	D	55
9	3.A.3.1; 3.B.1.3	3.A.3; 3.B.1	6.4; 1.5; 2.2	D	55
10	5.D.2.5	5.D.2	2.2	A	48
11	5.B.5.5	5.B.5	2.2; 6.4	D	51
12	1.B.1.1; 3.C.2.1	1.B.1; 3.C.2	6.4; 6.4	B	41
13	4.D.3.1	4.D.3	2.2	C	61
14	5.B.5.3	5.B.5	1.4; 2.2	B	49
15	5.D.3.1	5.D.3	6.4	B	38
16	2.B.1.1; 3.B.1.1; 3.B.2.1	2.B.1; 3.B.1; 3.B.2	2.2; 6.4; 1.1; 1.4; 2.2	A	38
17	4.B.2.2	4.B.2	5.1	B	37
18	3.E.1.3; 3.E.1.4; 4.C.2.1	3.E.1; 4.C.2	1.4; 2.2; 2.2; 6.4	B	40
19	5.B.9.3; 5.C.3.3	5.B.9; 5.C.3	2.2; 1.4; 2.2	B	49
20	5.C.3.1	5.C.3	6.4; 7.2	D	49
21	6.B.2.1	6.B.2	1.4	A	46
22	3.A.4.1	3.A.4	1.4; 6.2	D	44
23	4.C.1.1; 5.B.3.2	4.C.1; 5.B.3	1.4; 2.2; 1.4; 2.2	D	16
24	3.F.3.1	3.F.3	6.4; 7.2	D	40
25	5.B.3.3	5.B.3	1.4	A	28
26	3.B.1.3; 3.B.2.1	3.B.1; 3.B.2	1.5; 2.2; 1.1; 1.4; 2.2	B	48
27	6.B.1.1	6.B.1	1.4; 2.2	B	11
28	5.D.2.5	5.D.2	2.1	A	48
29	3.B.3.3	3.B.3	2.2	D	21
30	6.D.3.2; 6.D.4.2	6.D.3; 6.D.4	6.4; 2.2	D	69
31	3.F.1.1	3.F.1	1.4	B	27
32	3.A.1.1; 3.B.1.1	3.A.1; 3.B.1	1.5; 2.2; 6.4	B	46
33	3.C.1.2	3.C.1	2.2	B	27
34	4.A.1.1; 5.D.1.1; 5.D.2.1	4.A.1; 5.D.1; 5.D.2	1.2; 6.4; 6.4; 7.2	A	24
35	4.A.1.1	4.A.1	1.2; 1.4; 6.4	D	24
36	4.C.2.1; 5.B.5.5	4.C.2; 5.B.5	6.4; 2.2; 6.4	A	23

2017 AP Physics 1: Algebra-Based Question Descriptors and Performance Data

Question	Learning Objectives	Essential Knowledge	Science Practices	Key	% Correct
131	5.B.9.3	5.B.9	2.2	A,C	42
132	5.D.1.2	5.D.1	5.1; 5.3	C,D	43
133	3.A.3.1; 3.A.3.3; 3.A.4.2; 3.A.4.3	3.A.3; 3.A.4	6.4; 7.2; 1.4; 6.4; 7.2; 1.4	A,D	36
134	6.D.1.2	6.D.1	4.2	A,C	32

Free-Response Questions

Question	Learning Objectives	Essential Knowledge	Science Practices	Mean Score
1	5.B.9.3; 5.C.3.1; 5.C.3.3	5.B.9; 5.C.3	2.2; 6.4; 6.4; 1.4; 2.2	2.62
2	3.A.1.2; 3.A.1.3; 3.F.2.2; 3.F.3.3; 4.D.3.2	3.A.1; 3.F.2; 3.F.3; 4.D.3	4.2; 5.1; 4.1; 4.2; 5.1; 4.1; 4.2; 5.1; 4.1; 4.2	3.08
3	2.B.1.1; 3.A.1.1; 3.A.3.1; 3.B.1.1; 3.C.4.1; 4.C.1.1; 4.C.1.2; 4.C.2.1; 5.B.1.2; 5.B.3.1; 5.B.4.1; 5.B.4.2; 5.B.5.1; 5.B.5.4	2.B.1; 3.A.1; 3.A.3; 3.B.1; 3.C.1; 3.C.4; 4.C.1; 4.C.2; 5.B.1; 5.B.3; 5.B.4; 5.B.5	2.2; 7.2; 1.5; 2.2; 6.4; 6.4; 7.2; 6.1; 1.4; 2.1; 2.2; 6.4; 6.4; 1.5; 2.2; 6.4; 7.2; 6.4; 7.2; 1.4; 2.1; 2.2; 5.1; 6.4; 7.2	4.25
4	3.A.1.1; 3.A.4.1; 3.A.4.2; 3.B.1.1; 5.B.3.1; 5.B.3.3; 5.B.4.1; 5.B.4.2; 5.D.1.5; 5.D.2.1; 5.D.2.5	1.A.1; 3.A.1; 3.A.4; 3.B.1; 5.B.3; 5.B.4; 5.D.1; 5.D.2	1.5; 1.4; 6.2; 6.4; 6.4; 2.2; 6.4; 1.4; 2.2; 6.4; 1.4; 2.2; 2.2; 6.4; 2.1; 2.2	1.13
5	5.B.4.1; 6.A.1.2; 6.D.1.1	5.B.4; 6.A.1; 6.D.1	6.4; 7.2; 1.2; 1.1; 1.4	3.57

AP[®] PHYSICS 1
2017 SCORING GUIDELINES

Question 1

7 points total

**Distribution
of points**

(a) 4 points

For a correct ranking:

$$(I_3 = I_4) > I_1 > I_2 \text{ or } I_3 = I_4, I_1, I_2$$

1 point

Note: If an incorrect ranking is given, the explanation may still earn partial credit.

For indicating resistors in series must have the same current, and hence that $I_3 = I_4$

1 point

For explaining why I_1 and I_2 must be smaller than I_3 and I_4 in terms of current splitting at junctions or some other valid explanation

1 point

For explaining why $I_1 > I_2$ in terms of unequal resistances resulting in a current split with more current through the smallest resistance, OR equal potential drops resulting in $I_1 R_1 = I_2 R_2$ and hence $I_1 > I_2$

1 point

Example: $(I_3 = I_4) > I_1 > I_2$

All the current through the battery also goes through both R_3 and R_4 , and hence those two currents are the same. At the junctions to and from the parallel branch, the current “splits up” and passes through R_1 and R_2 before re-merging. So I_1 and I_2 are both less than I_3 and I_4 . Because R_2 has a greater resistance than R_1 , the current splits unevenly at the junction, with more current in the path with the least resistance. So the current in resistor R_1 will be greater than the current in R_2 .

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Question 1 (continued)

**Distribution
of points**

(b) 2 points

Correct answer: $\Delta V_2 = 1 \text{ V}$, $\Delta V_3 = 5 \text{ V}$

For recognizing that resistors in parallel have the same potential difference across them, and indicating $\Delta V_2 = 1 \text{ V}$

1 point

For applying the fact that $\Delta V = 0$ (explicitly or implicitly) for the loop containing the battery, R_1 or R_2 , and both R_3 and R_4 , and obtaining a value of ΔV_3 consistent with ΔV_2 (i.e. $\Delta V_2 + \Delta V_3 = 6 \text{ V}$)

1 point

Example: $\Delta V_2 = 1 \text{ V}$ $\Delta V_3 = 5 \text{ V}$

Since resistors R_1 and R_2 are in parallel, they have the same electric potential difference across them, so $\Delta V_2 = \Delta V_1 = 1 \text{ V}$. An electron that passes through the battery, R_2 , R_3 , R_4 , and back to the battery has completed one full loop around the circuit. Therefore, its overall potential change is $\Delta V = 0$. Therefore its potential change through the battery, of magnitude 15 V, equals the magnitude of its potential change across R_2 , R_3 and R_4 : $|\Delta V_2 + \Delta V_3 + \Delta V_4| = 15 \text{ V}$. So $\Delta V_3 = 15 \text{ V} - 9 \text{ V} - 1 \text{ V} = 5 \text{ V}$.

(c) 1 point

Answer: "Increase"

Note: Reasoning cannot earn credit if incorrect selection is made.

For recognizing that the addition of a wire provides a path of negligible resistance around R_3 and therefore reduces the equivalent resistance of the circuit, or other valid explanation.

1 point

Example: Adding a short circuit across R_3 gives the current a resistance-free path around R_3 . This lowers the overall resistance of the circuit, increasing the current through the battery and hence through R_1 .

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Question 2

12 points total

**Distribution
of points**

- (a) i and ii) 2 points
- For indicating valid measured quantities 1 point
- Note: This point is not earned if the procedure in part (iii) includes measured quantities or measurement equipment that is not listed in parts (i) and (ii).
- For a list of equipment and a plausible way to measure the quantities stated in (a)(i) 1 point
- iii) 3 points
- For a practical procedure consistent with the quantities indicated in (a)(i) 1 point
- For a plausible procedure that could be used to experimentally determine frictional torque 1 point
- For including a valid method to reduce experimental uncertainty 1 point
- Example methods:
- Method 1: Start the platform rotating and let it rotate freely. Use a feasible method for measuring the angular velocity at two different times.
- Method 2: Apply a known external force tangential to the rim of the initially motionless platform. Measure the change in angular velocity as in method 1. Also measure the radius of the platform (or some other valid measurement of lever arm).
- Method 3: Measure the torque needed to maintain the platform at a constant angular velocity. Determine if the torque is indeed constant by measuring the angular speed at two different times and seeing if they are the same within experimental uncertainty.
- Method 4: With the platform freely rotating, measure angular displacement over some time interval.
- (b) 3 points
- For a valid expression involving the necessary angular quantity consistent with the procedure in part (a) 1 point
- For explicitly or implicitly relating the calculated quantity to angular acceleration 1 point
- For correctly combining the above two concepts in a way that would yield the frictional torque 1 point
- (c) 2 points
- For correctly relating the procedure in parts (a) and (b) to the assumption of dependence or independence of frictional torque on angular speed 1 point
- For connecting the analysis to the dependence of frictional torque on angular speed 1 point
- (d) 2 points
- For a modification that has any additional trial at a different angular speed OR correctly and explicitly stating that no modification is necessary 1 point
- For indicating how analysis of the initial or modified experiment could plausibly determine whether frictional torque is independent of angular speed 1 point

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Question 3

12 points total

**Distribution
of points**

(a) 2 points

For indicating (explicitly or implicitly) that intersecting the vertical axis corresponds to zero stiffness 1 point

For correctly indicating that no measurement of stiffness could be zero 1 point

Example: The vertical axis is where $S = 0$, indicating a substance puts up no resistance to getting stirred. But even at room temperature the oil has *some* stiffness. Hence, the lowest measured stiffness is positive, not zero, and the graph reflects this.

(b) 3 points

For a correct equation: $E_{\max} = MgD$ OR $E_{\max} = -MgD$ 1 point

For stating that the maximum energy that can be lost is equal to the initial energy of the system 1 point

For stating that the initial energy of the system is the initial gravitational potential energy 1 point

Example: The maximum energy that can be lost is equal to the initial energy, which is all gravitational potential energy: MgD .

Alternate solution for the last 2 points

For a statement or the application of mechanical energy as the sum of kinetic and potential energies 1 point

For combining the change in the gravitational potential energy with a small change in kinetic energy as the ball falls to make a valid argument for $E_{\max}$. 1 point

Example: In a very stiff oil that the ball can barely fall through, it gains essentially no kinetic energy. In this case the overall change in mechanical energy, $\Delta K + \Delta U$, is approximately equal to ΔU because ΔK is essentially zero.

So, $E_{\max} = |\Delta U| = MgD$.

(c) 2 points

For each of the following, up to a maximum of 2 points

1 point for each valid, independent statement that the equation does not match the graph (maximum of 2 points).

1 point for a statement that demonstrates that E_{lost} cannot increase indefinitely.

2 points

Example statements (1 point each, any two could be combined for 2 points):

The equation indicates that E_{lost} increases indefinitely with S , but the graph has an upper limit for E_{lost} .

The equation represents a parabola (or is concave-up), which does not match the graph.

It doesn't make physical sense that E_{lost} could be more than the available potential energy MgD .

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Question 3 (continued)

**Distribution
of points**

(d) 2 points

Correct answer: “Not plausible”

Note: The explanation can earn the first point if the incorrect selection is made.

For a statement or implication that the equation indicates that t increases as S decreases

1 point

For a statement or implication that t should be larger as S increases

1 point

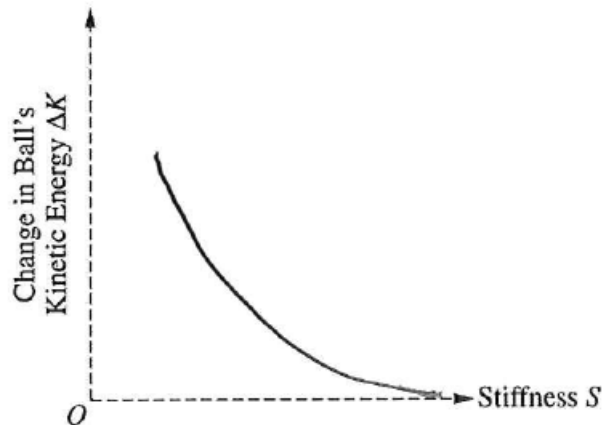
Note: If only one of the above statements is written explicitly, and “Not plausible” is selected, the second point is earned by implication.

Example 1 (2 points): “Not plausible”. According to the equation, a less stiff oil makes the sphere take more time to reach the bottom.

Example 2 (2 points): “Not plausible”. A greater stiffness means that the ball should take a longer time to fall through the oil.

Example 3 (1 point): “Plausible”. The equation says that bigger S means bigger time, which agrees with the idea that a thicker oil resists the ball’s motion more.

(e) 3 points



For drawing a concave-upward curve that approaches the horizontal at higher S

1 point

For having the maximum value of ΔK at the lowest graphed value of S , which can be $S = 0$ or a positive value

1 point

Note: This point is not earned if the graph indicates that ΔK approaches infinity as S approaches zero.

For high values of S , the graph approaches and does not cross the horizontal axis

1 point

Note: This point is earned if the graph touches the horizontal axis without crossing it.

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Question 4

7 points total

**Distribution
of points**

(a)

Correct answer: 2 θ_{embed} 3 θ_{pass} 1 θ_{bounce}

(b) 7 points

- | | |
|--|---------|
| For using conservation of momentum | 1 point |
| Note: This point is earned for an indication of how momentum conservation applies to the situation. | |
| For correctly stating or implying that in all three cases the magnitude of the change in the horizontal momentum of the arrow during the collision equals magnitude of the momentum gained by the pumpkin | 1 point |
| For a valid argument explaining why the Bounce arrow must transfer the most momentum | 1 point |
| For stating that the Pass arrow ends up faster than the Embedded arrow or does not transfer as much momentum to the pumpkin immediately after collision | 1 point |
| For indicating (regardless of conclusion about whether the Pass or Embedded arrow ends up going faster after the collision) that the arrow that slowed more during the collision transfers more momentum to the pumpkin | 1 point |
| For an explicit or implicit correct justification, in terms of energy conservation, that the angle reached by the pumpkin is related to the speed of the pumpkin just after the collision | 1 point |
| For a logical, relevant, and internally consistent response that addresses the required argument or question asked, and follows the guidelines described in the published requirements for the paragraph-length response | 1 point |

Example: The Bounce arrow is the only one that reverses direction, so has momentum to the left after the collision. By conservation of momentum the pumpkin must then have a greater momentum to the right than the arrow had before the collision. After the Embedded arrow hits, the arrow and pumpkin together have the same momentum as the arrow did just before the collision. The Pass arrow still has some momentum to the right after hitting the pumpkin so by conservation of momentum the pumpkin has less momentum to the right than the arrow did before the collision. By conservation of energy, the maximum angle that the pumpkin reaches depends on the speed of the pumpkin (higher speed means higher angle), so the Bounce arrow pumpkin has the highest angle and the Pass arrow pumpkin has the lowest angle.

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Question 5

7 points total

**Distribution
of points**

(a) 1 point

Answer: “Yes”

Explanation cannot earn credit if “No” is selected.

For a statement of conservation of energy, or explicit reference to the motion or kinetic energy of the string at the moment shown in the diagram

1 point

(b) 1 point

Correct Answer: “Neither”

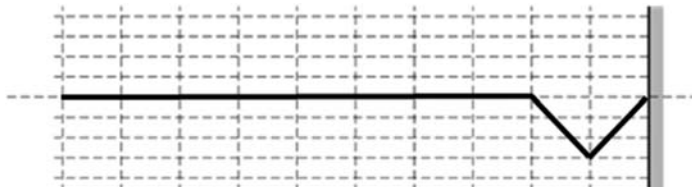
Explanation cannot earn credit if either of the other choices are selected.

For an explanation that both pulses travel at the same speed because wave speed is a property of the string, which does not change

1 point

(c)

i) 3 points



For drawing the entire wave pulse below the equilibrium position of the string

1 point

For drawing a wave pulse with amplitude of -3 units

1 point

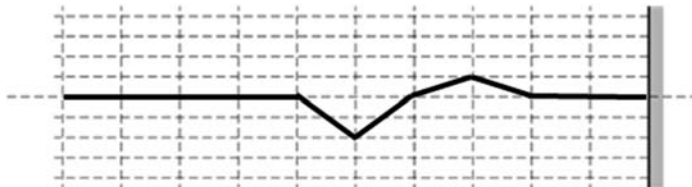
Note: This point is not earned if the equilibrium position or left-most portion of the pulse is drawn or indicated at the incorrect height.

For a pulse with symmetric triangular shape that is 2 units wide and located in the rightmost 2 units of the grid, and no other nonzero portions of the string

1 point

Note: The portion of the string with zero displacement need not be sketched.

ii) 2 points



For a pulse that is a vertical inversion of the original pulse (i.e. 2 unit portion going down and 1 unit portion going up)

1 point

For a pulse that is a horizontal inversion of the original pulse (i.e. 2 unit portion on the left and 1 unit portion on the right)

1 point

Note: The portion of the string with zero displacement need not be sketched.